

(a) Fix weights; adapt counts

sets task importance

Task weights w_i

fixed 



controls estimate precision

Micro-batches per task m_i

adaptive



current $m = (4, 4, 4, 4)$; 16 micro-batches total

task- i micro-batch loss $\times w_i/m_i$

$$\text{step gradient: } A = \sum_i \frac{w_i}{m_i} \sum_{s=1} g_{i,s}$$

same expected gradient for every m

(b) Measure noise and time

from the completed optimizer step

micro-batch gradient $g_{i,s}$

i : task; s : micro-batch within task i

AdamW-scaled gradient $Dg_{i,s}$

D : frozen diagonal scale; optimizer unchanged

within-task noise estimate $\hat{e}_i$

sample variance of $Dg_{i,s}$ over s



τ_i : time per micro-batch for task i

C : fixed overhead per optimizer step

(c) Choose next counts

inputs

weights w_i

noise $\hat{e}_i$

Cost only: τ_i, C

GPAS

allocate by $w_i \sqrt{\hat{e}_i}$

minimize $\sum_i w_i^2 \hat{e}_i / m_i$ per step

Cost-GPAS

minimize time $\times$ variance

best variance at a fixed wall-clock budget

next $m = (3, 6, 2, 5)$

16 total; 2-8 per task



next step: use new counts; keep task weights fixed